

IBM Data Engineer Complete Preparation Roadmap

This handbook provides a structured roadmap for preparing for IBM Data Engineer interviews.

1. Programming Fundamentals

Python, SQL, Linux, Git.

2. Python Practice

Reverse String, Fibonacci, Prime, Armstrong, Factorial, GCD/LCM, Dictionaries, CSV, File Handling, Exceptions, Decorators, Generators, Lambda, Map, Filter, Reduce.

3. SQL Practice

JOINS, GROUP BY, HAVING, CTEs, Window Functions, 100+ LeetCode SQL questions.

4. DSA

Arrays, Strings, Hash Maps, Stacks, Queues, Linked Lists, Trees, Graphs, Dynamic Programming, Binary Search.

5. Big Data

Apache Spark, PySpark, Hadoop, Hive, Kafka, Airflow.

6. Cloud

IBM Cloud, AWS basics, Azure basics.

7. Data Engineering

ETL, ELT, Data Warehousing, Data Lakes, Partitioning, Data Modeling.

8. Projects

5 end-to-end projects using Python, SQL, Spark, Airflow, Kafka.

9. Interview Preparation

Coding rounds, SQL rounds, system design, behavioral questions.

10. Practice Links

LeetCode: <https://leetcode.com/problemset/>

LeetCode SQL: <https://leetcode.com/problemset/database/>

HackerRank Algorithms: <https://www.hackerrank.com/domains/algorithms>

HackerRank SQL: <https://www.hackerrank.com/domains/sql>

HackerRank Python: <https://www.hackerrank.com/domains/tutorials/10-days-of-python>

StrataScratch: <https://www.stratascratch.com/>